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# Atomic Connectivity and Spatial Group Contribution Models for Accurate Prediction of Saturated Vapor Pressure with Boiling-Point

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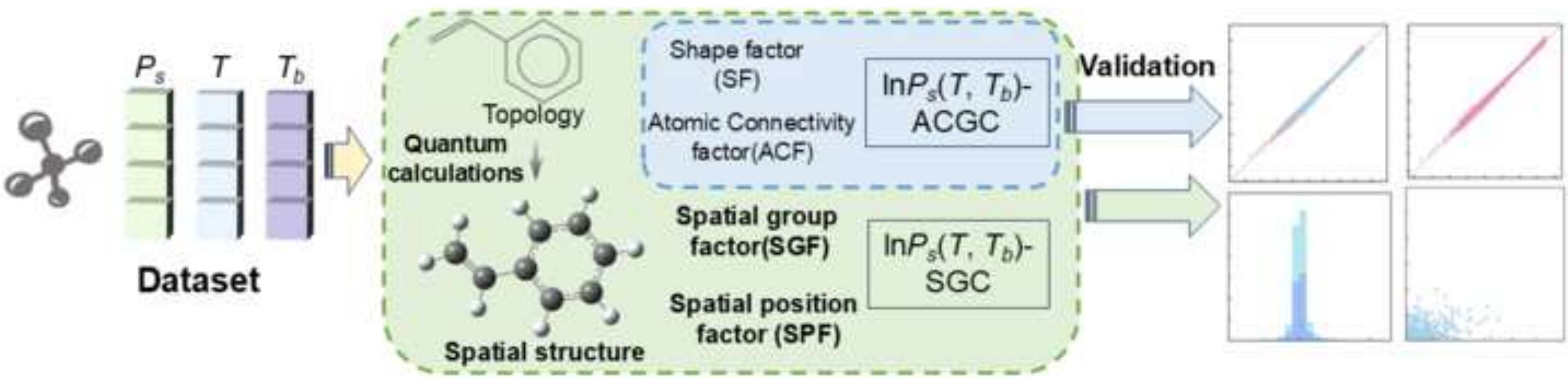
| Abstract: | Saturated vapor pressure (Ps) is a crucial thermodynamic property in the chemical engineering, playing an irreplaceable role in phase equilibrium calculations, separation process design, and mass transfer analysis. In this study, the Atomic Connectivity Group Contribution (ACGC) and Spatial Group Contribution (SGC) models are developed for predicting Ps by introducing the normal boiling point (Tb). During the modeling process, the ACGC method primarily considers the topological structure of molecules. Given the stereochemical characteristics of molecular configurations, the properties exhibited by the same functional group can vary depending on its spatial position or conformation within a molecule. The SGC method introducing the Spatial Group Factor (SGF) and Spatial Position Factor (SPF) is further applied to quantify the spatial structure of molecules. The lnPs(T, Tb)-ACGC and lnPs(T, Tb)-SGC models achieve average absolute error (AAE) of 0.067 and 0.056 ln(kPa) for test set, respectively. Through internal validation, both models demonstrate excellent stability (Q2LOO: 0.9972 and 0.9981). The research results indicate that the lnPs(T, Tb)-SGC model shows better performance than the lnPs(T, Tb)-ACGC model in terms of Ps prediction accuracy. The lnPs(T, Tb)-ACGC model employs a strategy based on molecular shape factors and atomic connectivity factor, maintaining a relatively simple architecture while achieving efficient property estimation. In contrast, the lnPs(T, Tb)-SGC model demonstrates enhanced predictive performance through full consideration of spatial structural characteristics, albeit at the expense of increased model complexity. |



$P_s$
$T$
$T_b$
Dataset
Topology
Quantum calculations
Spatial structure
Shape factor (SF)
Atomic Connectivity factor(ACF)
$\ln P_s(T, T_b)$-ACGC
Spatial group factor(SGF)
Spatial position factor (SPF)
$\ln P_s(T, T_b)$-SGC
Validation

# Atomic Connectivity and Spatial Group Contribution Models for Accurate Prediction of Saturated Vapor Pressure with Boiling-Point

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## Abstract

Saturated vapor pressure ($P_s$) is a crucial thermodynamic property in the chemical engineering, playing an irreplaceable role in phase equilibrium calculations, separation process design, and mass transfer analysis. In this study, the Atomic Connectivity Group Contribution (ACGC) and Spatial Group Contribution (SGC) models are developed for predicting $P_s$ by introducing the normal boiling point ($T_{\mathrm{b}}$). During the modeling process, the ACGC method primarily considers the topological structure of molecules. Given the stereochemical characteristics of molecular configurations, the properties exhibited by the same functional group can vary depending on its spatial position or conformation within a molecule. The SGC method introducing the Spatial Group Factor (SGF) and Spatial Position Factor (SPF) is further applied to quantify the spatial structure of molecules. The ln$P_s(T, T_b)$-ACGC and ln$P_s(T, T_b)$-SGC models achieve average absolute error (AAE) of 0.067 and 0.056 ln(kPa) for test set, respectively. Through internal validation, both models demonstrate excellent stability ($Q^2_{\mathrm{LOO}}$: 0.9972 and 0.9981). The research results indicate that the ln$P_s(T, T_b)$-SGC model shows better performance than the ln$P_s(T, T_b)$-ACGC model in terms of $P_s$ prediction accuracy. The ln$P_s(T, T_b)$-ACGC model employs a strategy based on molecular shape factors and atomic connectivity factor, maintaining a relatively simple architecture while achieving efficient property estimation. In contrast, the ln$P_s(T, T_b)$-SGC model demonstrates enhanced predictive performance through full consideration of spatial structural characteristics, albeit at the expense of increased model complexity.

**Keywords:** Group contribution (GC), Atomic Connectivity Group Contribution (ACGC), Spatial group contribution (SGC), Saturated vapor pressure ($P_s$), Normal boiling point ($T_b$)

## 1. Introduction

At a certain temperature, the evaporation and condensation rates between the liquid and the gas molecules are equal, and the gas no longer evaporates or condenses, at which point the whole system reaches an equilibrium state, and the corresponding pressure of the gas phase is the saturated vapor pressure ($P_s$). The $P_s$ is an important thermodynamic property that describes the equilibrium state of the gaseous and liquid phases and is essential in phase equilibrium calculations[1-4]. This property can also quickly determine the volatility, air-water partition coefficient, and migration characteristics of chemicals, and plays an important role in chemical process design (e.g., distillation process), simulation, and contaminant transport[5-9]. The $P_s$ is also correlated with surface tension[10-12], enthalpy of vaporization[13, 14], Henry's law constant[15], boiling point[16] and other physicochemical properties, which are widely used. In addition, in the context of global warming, the design and screening of refrigerants has become the focus, which has also led to more attention to the knowledge of $P_s$ [17-20].

In the early stages, researchers primarily obtained $P_s$ data for different substances through experimental measurements. However, this approach is often time-consuming and costly, particularly for newly synthesized compounds or measurements under extreme temperature conditions. To address this limitation, various vapor pressure estimation models were subsequently developed. The earliest Clapeyron equation (1834) laid the theoretical foundation, followed by a series of models such as the Antoine equation[21], Wagner equation[22], and Riedel equation[23], Ambrose-Walton equation[24]. Although these correlation-state-principle-based formulas perform well under specific conditions, they still exhibit notable limitations, for example significant deviations near the critical temperature or triple point and frequent requirement of excessive adjustable parameters[25]. Among them, the most widely used Antoine equation[21], despite its simplicity, suffers from pronounced constraints in prediction accuracy due to temperature range and substance category dependencies[26, 27]. Additionally, researchers attempted to optimize the models by incorporating key parameters such as the normal boiling point ($T_b$) [28-30]. For instance, Roosta and Hekayati[25] employed a genetic programming to screen thermophysical constants and developed a simplified $P_s$ expression based on $T_b$ and molecular weight. Ohe[2] then developed a prediction scheme for vapor pressure directly determined by boiling point based on the Clapeyron equation and the Antoine equation, achieving average absolute deviations of 0.709 kPa and 0.615 kPa, respectively.

With the advancement of modeling approaches, various theoretical and semi-empirical models have emerged to predict and estimate vapor pressure based on molecular structure and thermodynamic properties. These methods, which derive property values from molecular structure, can generally be classified into quantitative structure-property relationship (QSPR)[31] and group contribution (GC)[32-34] methods. The core idea involves quantifying molecular structure, where the contribution of each structural parameter is individually considered, and then these contributions are summed to estimate the overall vapor pressure of the molecule. Godavarthy et al.[35] have developed a universal the scaled variable reduced coordinates (SVRC)-QSPR model based on 52445 $P_s$ data points from 1221 compounds, which can accurately predict $P_s$ across the range from the triple point to the critical point. Gharagheizi et al.[5, 15] established two vapor pressure models at different temperatures using QSPR and GC methods, achieving squared correlation coefficients ($R^2$) of 0.990 and 0.994, respectively.

In another study, Neaf and Acree[36] developed a predictive model using the GC method, incorporating 1907 vapor pressure data points. Although the model demonstrates good predictive accuracy and stability, its applicability is somewhat limited due to the collection of vapor pressure data only at 298.15 K. Recently, Feng et al.[31] developed two QSPR models using norm descriptors, with one using temperature ($T$) as the known parameter ($R^2$=0.9970) and another one incorporating both $T$ and $T_b$ as known parameters ($R^2$=0.9975). The results demonstrate that introducing the $T_b$ parameter significantly improved the predictive accuracy of model.

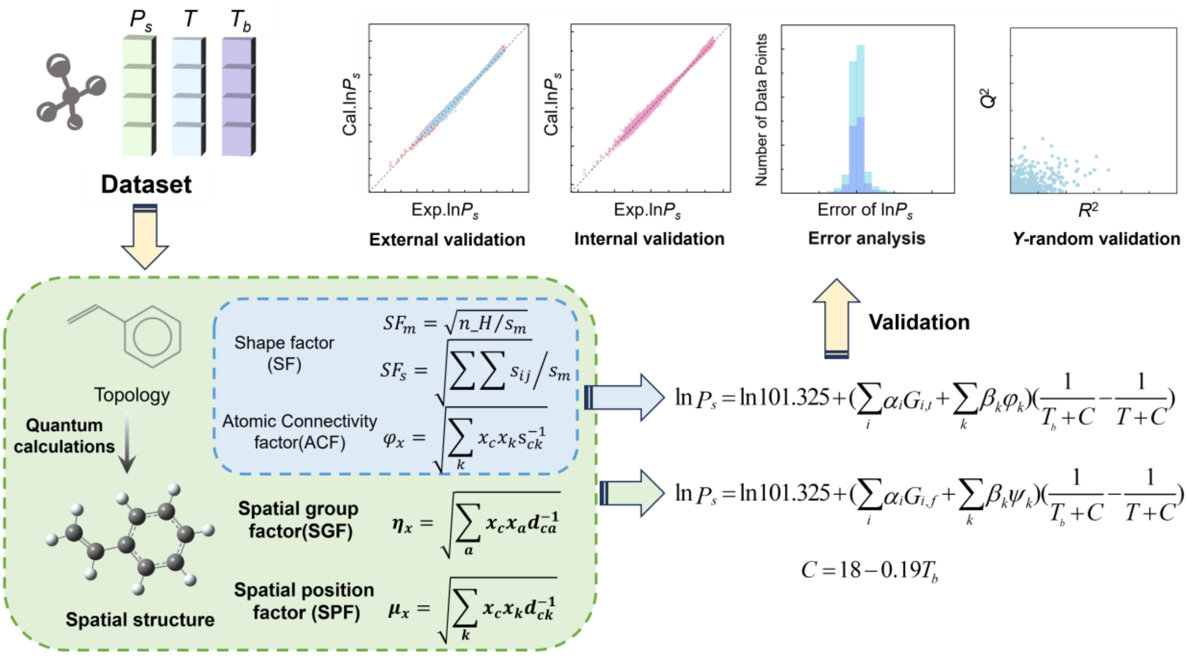

Figure 1. Schematic of the model development process

The Atomic Connectivity Group Contribution (ACGC)[37] and Spatial Group Contribution (SGC)[38] methods have been shown their strengths in predicting the properties (e.g., critical temperature ($T_c$), critical pressure ($P_c$), critical volume ($V_c$), boiling point ($T_b$), and melting point ($T_m$)) of organic compounds by defining the group factor and the position factor. In this study, two GC models are developed by introducing the easily measured $T_b$ into

ACGC and SGC methods for predicting saturated vapor pressure ($P_s$), as shown in Figure 1. Compared with the traditional GC method, the ACGC models developed in this study introduce interatomic step lengths and fundamental atomic properties to achieve quantitative characterization of shape factors (SFs) and atomic connectivity factors (ACFs). On the other hand, the SGC method integrates interatomic Euclidean distances and atomic quantum chemical properties to mathematically quantify group spatial configuration (Spatial Group Factor-SGF) and positional information (Spatial Position Factor-SPF), thereby replacing the simple summation of group numbers in traditional methods. The predictive capability, stability, and reliability of the two $P_s$ models are evaluated through external validation, internal validation, and $Y$-random validation methods.

## 2. Methodology

### 2.1. Data sets

The $P_s$ data in this work are from Feng et al.[31], while requiring that the functional groups involved must appear in at more than five compounds to ensure sufficient data for model training. Through the systematic screening process, the dual-parameter modeling group (incorporating both $T$ and $T_b$) comprise 17800 $P_s$ data points representing 1062 compounds.

For dataset partitioning, we adopt the following principles: (1) ensuring each functional group appears at least five times in the training set to meet basic data requirements for model training; (2) implementing strict compound-based division to guarantee complete separation between training and test sets. Ultimately, the compounds and their corresponding data points are allocated in a 4:1 ratio, achieving a rational split between training and test sets. The specific partitioning results are presented in Table 1 and the statistics of common families of molecules and the temperature distribution of data points are illustrated in Figure 2. Throughout the dataset, there is a wide range of $T$, ranging from as low as 170 K to as high as 974 K. The $T_b$ range is similarly broad, ranging between 267.09 and 725.3 K.

Table 1. The data segmentation results

| Samples | | Dataset | Train set | Test set | The type of group |
|---|---|---|---|---|---|
| $\ln P_s(T, T_b)$ | Compounds | 1062 | 850 | 212 | 77 |
| | Datapoints | 17800 | 14237 | 3563 | |
| | $T$ (K) | 170~974 | 175.99~974 | 170~797 | |
| | $T_b$ (K) | 267.09~725.3 | 267.09~725.3 | 272.76~724.49 | |

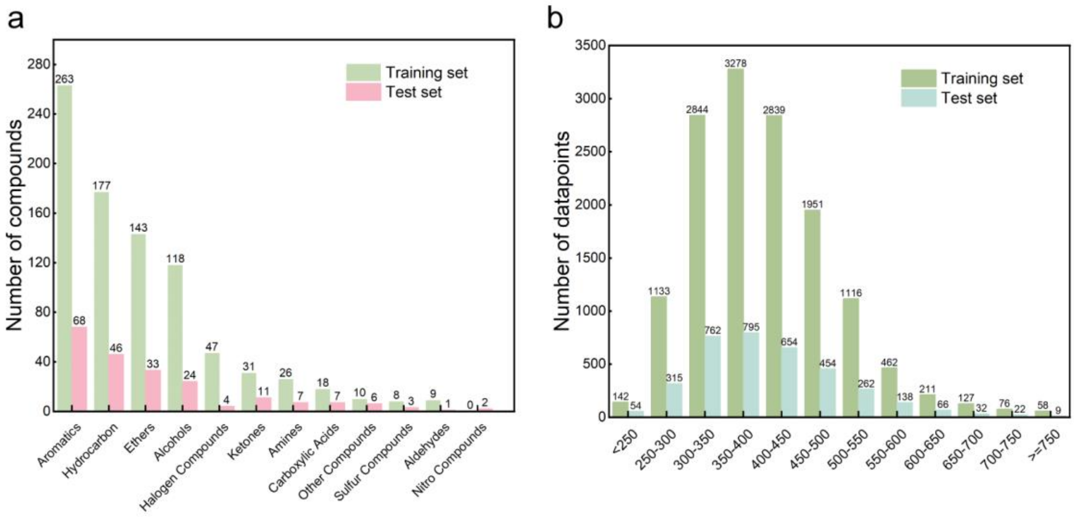


Figure 2. The information of database

## 2.2. Group division

The division of groups is based on the Atomic Adjacent Group (AAG), which classifies atoms according to the connection relationship between atoms, distinguishing between connection atoms and endpoint atoms. Groups are represented in the order of core atoms, endpoint atoms, and connection atoms, for example: C(-H)[∵C][∵C]. Here, '-' represents a straight-chain single bond, and '∵' denotes an aromatic bond.

## 2.3. Topological parameter calculation

This work first extracts features from the molecular structures, including interatomic step length information ($S$) and atomic intrinsic properties ($P$). Then, these features are integrated through specific mathematical formulas (Eqs. (1)-(3)), ultimately the Shape Factor (SF) and the Atomic Connectivity Factor (ACF) of groups are calculated.

$$SF_m = \sqrt{n\_H / s_m} \tag{1}$$

$$SF_s = \sqrt{\sum_i \sum_j s_{ij}} \Big/ s_m \tag{2}$$

$$\varphi_x = \sqrt{\sum_k x_c x_k s_{ck}^{-1}} \tag{3}$$

where, $n\_H$ is the number of non-hydrogen atoms; $s_m$ is the maximum step of atoms from H-suppressed structure; $s_{ij}$ is the steps between non-hydrogen atoms $i$ and $j$. The $s_{ck}$ is the steps between core atom $c$ and other non-hydrogen atom $k$; $x$ stands for atomic properties. $\varphi_k$ involves 7 atomic properties, including atomic weight (*wei*), electronegativity (*ele*), ionization energy (*ion*), atom radius (*rad*), the number of outermost electrons (*noe*), the number of outermost electrons divided by the number of electron shells (*noe/nes*), and branched degree (*bra*).

## 2.4. Spatial parameter calculation

To realize the spatial differentiation characteristics of molecular structures, geometric optimization calculations at the B3LYP functional and 3-21G basis set level are performed using Gaussian software[39], systematically obtaining the interatomic distance information ($D$) and quantum chemical information of atoms ($Q$). On this basis, two quantitative indicators - the Spatial Group Factor (SGF) and the Spatial Position Factor (SPF) - are calculated to characterize the three-dimensional spatial configuration and spatial position of molecular groups, respectively. The specific mathematical expressions are given in Eqs. (4) and (5).

$$\eta_x = \sqrt{\sum_a x_c x_a d_{ca}^{-1}} \tag{4}$$

$$\mu_x = \sqrt{\sum_k x_c x_k d_{ck}^{-1}} \tag{5}$$

where, $d_{ca}$ is the distance between the core atom $c$ and its adjacent non-hydrogen atoms $a$; $d_{ck}$ is the distance between core atom $c$ and other non-hydrogen atoms $k$; the properties of hydrogen atoms are superimposed on the atoms connected to hydrogen; $x$ stands for the quantum properties, mainly including the Mulliken charges ($Mc$), the charge for natural population analysis (NPA) ($cn$), the number of core electrons for NPA ($ncn$), the number of valence electrons for NPA ($nvn$), the number of Rydberg electrons for NPA ($nRn$), the number of total electrons for NPA ($ntn$), the energy of core electrons for NPA ($ecn$), the energy of valence electrons for NPA ($evn$), the energy of Rydberg electrons for NPA ($eRn$), the energy of total electrons for NPA ($etn$), the gross orbital population of core electrons ($gc$), the gross orbital population of valence electrons ($gv$), the gross orbital population of Rydberg electrons ($gR$), and the gross orbital population of total electrons ($gt$).

## 2.5. Model development

The SGC model consists of group values ($G_{i,f}$) and position factors ($\psi_k$). The $P_s$ of a compound is typically described by the Antoine equation (Eq. (6)), where the $T$ at a $P_s$ of 101.325 kPa corresponds to the $T_b$ of the substance. In this study, the SGC method is used to develop $P_s$ models with $T$ and $T_b$ as parameters (Eq. (7)). To simplify the estimation of $P_s$, the model based on the ACGC method is also established as a supplement (Eq. (8)). Parameter $C$[40] exhibits a linear relationship with the $T_b$, as expressed in Eq. (9). During the model construction process, Bidirectional Stepwise Regression is adopted to screen key group parameters, followed by linear regression analysis using the Least Squares method.

$$\ln P = A - \frac{B}{T+C} \tag{6}$$

$$\ln P_s = \ln 101.325 + \left(\sum_i \alpha_i G_{i,f} + \sum_k \beta_k \psi_k\right)\left(\frac{1}{T_b + C} - \frac{1}{T+C}\right) \tag{7}$$

$$\ln P_s = \ln 101.325 + (\sum_i \alpha_i G_{i,t} + \sum_k \beta_k \varphi_k)(\frac{1}{T_b + C} - \frac{1}{T + C}) \tag{8}$$

$$C = 18 - 0.19 T_b \tag{9}$$

where, $\alpha_i$ and $\beta_k$ are adjustable parameters. $G_{i,f}$ is composed of $G_i$, $G_i/SF_m$, $G_i/SF_s$, and $\eta_i$, while $G_{i,t}$ consists of $G_i$, $G_i/SF_m$, and $G_i/SF_s$. $G_i$ represents the number of groups; $SF_m$ and $SF_s$ are shape factors; and $\eta_i$ denotes the spatial configuration of the group. During model fitting, each group selects one $G_{i,f}$ from $G_i$, $G_i/SF_m$, $G_i/SF_s$, and $\eta_i$ to participate in model development ($G_{i,t}$ is similar). $\psi_k$ is formed by $\varphi_k$ and $\mu_k$. $\varphi_k$, the atomic connectivity factor, incorporates seven atomic properties and characterizes the topological position of group. $\mu_k$ describes the spatial position of group and involves 14 quantum properties. In the SGC method (Eq. (7)), each group selects no more than one $\psi_k$ from 21 position factors. Similarly, In the ACGC method (Eq. (8)), each group selects no more than one $\varphi_k$ from 7 position factors. The parametric results of the $\ln P_s(T, T_b)$-ACGC and $\ln P_s(T, T_b)$-SGC models are shown in Tables S1 and S2.

### 2.6. Model evaluation

External validation, internal validation, and $Y$-random validation are the most widely used comprehensive validation methods in the modeling process. External validation involves testing the predictive performance of an established model using a test set, where the difference between the experimental and predicted values of the test set reflects the predictive capability of model. Internal validation is typically employed to assess the stability and robustness of the model, with the leave-one-out cross-validation (LOO-CV) method being the most common. LOO-CV is a rigorous model evaluation technique, and the process is as follows: for each substance in the training set, the study selects it individually as the test set while using all remaining substances as the training set to reconstruct the model. Subsequently, this newly constructed model is used to predict the previously isolated test set substance. This step is repeated for each substance in the training set, ensuring that every substance can be independently evaluated as a test set. Finally, each substance in the training set has a new predicted result, and the difference between its predicted and experimental values can reflect the stability of model. The chance correlation of the model is evaluated through 1000 of *Y*-random validation. This validation involves randomly shuffling the experimental values 1000 times and then comparing them with the calculated values to demonstrate the absence of chance correlations in the modeling process.

In addition, statistical parameters, including squared correlation coefficient ($R^2$), squared correlation coefficient for LOO-CV ($Q^2_{\mathrm{LOO}}$), average absolute error (AAE), as provided in Eqs. (10)-(12), are used to evaluate the goodness of fit of the models.

$$R^2 = \frac{\left(\sum\left(Y_{i,\mathrm{exp}} - \overline{Y}_{\mathrm{exp}}\right)\left(Y_{i,\mathrm{pred}} - \overline{Y}_{\mathrm{pred}}\right)\right)^2}{\sum\left(Y_{i,\mathrm{exp}} - \overline{Y}_{\mathrm{exp}}\right)^2 \sum\left(Y_{i,\mathrm{pred}} - \overline{Y}_{\mathrm{pred}}\right)^2} \tag{10}$$

$$Q^2_{\mathrm{LOO}} = \frac{\left(\sum\left(Y_{i,\mathrm{exp}} - \overline{Y}_{\mathrm{exp}}\right)\left(Y_{i,\mathrm{LOO\text{-}pred}} - \overline{Y}_{\mathrm{LOO\text{-}pred}}\right)\right)^2}{\sum\left(Y_{i,\mathrm{exp}} - \overline{Y}_{\mathrm{exp}}\right)^2 \sum\left(Y_{i,\mathrm{LOO\text{-}pred}} - \overline{Y}_{\mathrm{LOO\text{-}pred}}\right)^2} \tag{11}$$

$$\mathrm{AAE} = \frac{\sum\left|Y_{i,\mathrm{exp}} - Y_{i,\mathrm{pred}}\right|}{N} \tag{12}$$

where, $Y_{i,\mathrm{exp}}$ and $Y_{i,\mathrm{pred}}$ are the experimental and predicted values of the property *Y* for the compound *i*, respectively. $Y_{i,\mathrm{LOO\text{-}pred}}$ describes the predicted value of LOO-CV, *N* is the number of data points.

## 3. Results and discussion

Table 2 summarizes the predictive performance of the $\ln P_s(T, T_b)$-ACGC model versus the $\ln P_s(T, T_b)$-SGC model. The data demonstrate that the $\ln P_s(T, T_b)$-SGC model exhibits superior fitting capability with $R^2_{\mathrm{training}}$ of 0.9987, showing a slight improvement over that of

0.9982 achieved by the ln$P_s$($T$, $T_b$)-ACGC model. In terms of prediction accuracy, the $AAE_{training}$ of the SGC model decreases to 0.057 ln(kPa) from 0.067 ln(kPa) in the ACGC model, representing a significant reduction of 14.93%. These results indicate that the SGC approach incorporating molecular spatial structural features can more effectively capture the intrinsic patterns of molecular properties, thereby substantially enhancing the reliability of model predictions. In addition, the ACGC method that only considers the topological properties can also achieve sufficient high precision in prediction.

Table 2. The results of the ln$P_s$($T$, $T_b$)-ACGC model versus the ln$P_s$($T$, $T_b$)-SGC model

| Models | Methods | $R^2$/$Q^2_{LOO}$ | AAE[a] | $AE_{max}$[a] | $AE_{min}$[a] | Variables |
|---|---|---|---|---|---|---|
| ln$P_s$($T$, $T_b$)-ACGC | Training set | 0.9982 | 0.067 | 1.23 | $1.90\times10^{-6}$ | 77[b]+26[c] |
| | Test set | 0.9982 | 0.067 | 1.37 | $5.92\times10^{-7}$ | |
| | LOO-CV | 0.9972 | 0.081 | 2.19 | $1.96\times10^{-6}$ | |
| | Total | 0.9982 | 0.067 | 1.37 | $5.92\times10^{-7}$ | |
| ln$P_s$($T$, $T_b$)-SGC | Training set | 0.9987 | 0.057 | 0.96 | $9.32\times10^{-7}$ | 77[d]+28[e] |
| | Test set | 0.9987 | 0.056 | 1.39 | $7.10\times10^{-6}$ | |
| | LOO-CV | 0.9981 | 0.067 | 1.29 | $1.12\times10^{-6}$ | |
| | Total | 0.9987 | 0.057 | 1.39 | $9.32\times10^{-7}$ | |

[a]: ln(kPa), [b]: The number of groups ($G_{i,t}$), [c]: The number of position factors ($\varphi_k$), [d]: The number of groups ($G_{i,f}$), [e]: The number of position factors ($\psi_k$).

### 3.1. Validations of model performances

Figures 3 and 4 present the validation results of the ln$P_s$($T$, $T_b$)-ACGC and ln$P_s$($T$, $T_b$)-SGC models, respectively. A comparative analysis reveals that the ln$P_s$($T$, $T_b$)-ACGC ($R^2_{test}$=0.9982，$Q^2_{LOO}$=0.9972) and ln$P_s$($T$, $T_b$)-SGC models ($R^2_{test}$=0.9987，$Q^2_{LOO}$=0.9981) exhibit a tightly clustered distribution around the diagonal line (y = x) in both external and internal validations. Furthermore, the absolute errors (AE) of LOO-CV for the ln$P_s$($T$, $T_b$)-ACGC model range from $1.96\times10^{-6}$ ln(kPa) to 2.19 ln(kPa), whereas the ln$P_s$($T$, $T_b$)-SGC model demonstrates a significantly narrower error range of only $1.12\times10^{-6}$ ln(kPa) to 1.29 ln(kPa). These findings indicate that the ln$P_s$($T$, $T_b$)-SGC model not only delivers superior predictive performance but also exhibits enhanced stability. The ACGC method also achieves satisfactory prediction performance and stability through simple calculations. Additionally, the $Y$-random validation results (Figures 3d and 4d) for both models yield values substantially lower than their respective $R^2_{\text{training}}$ and $Q^2_{\text{LOO}}$ metrics, confirming the absence of chance correlation during model developments.

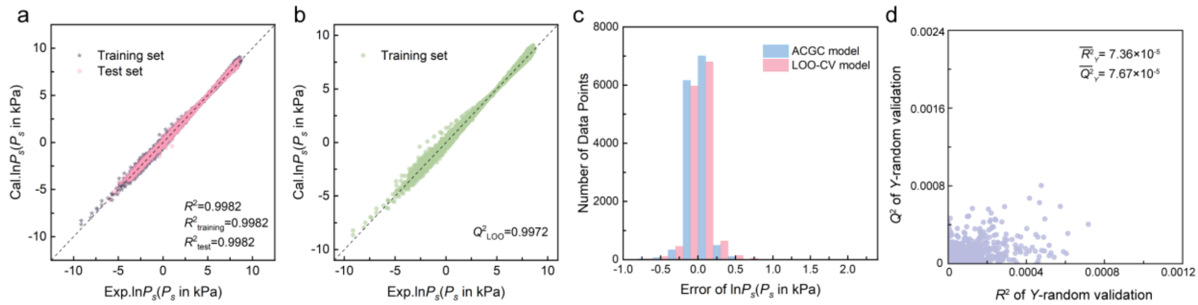


Figure 3. Validation results of the ln$P_s$($T$, $T_b$)-ACGC model: (a) external validation, (b) internal validation, (c) error distributions of the LOO-CV and ACGC models, and (d) $Y$-random validation

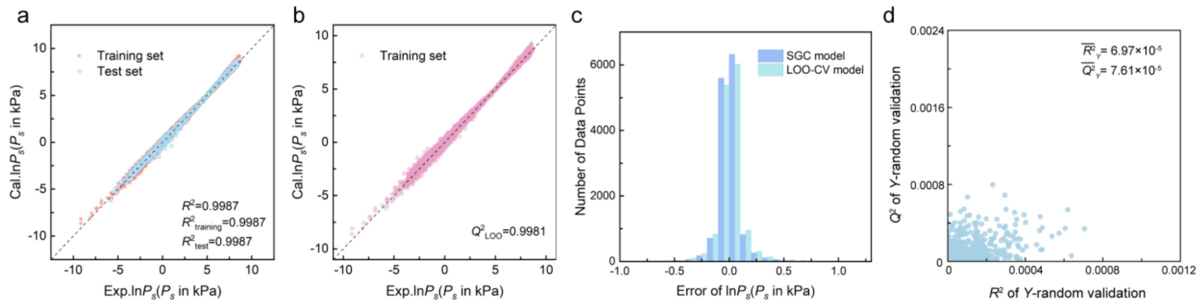


Figure 4. Validation results of the ln$P_s$($T$, $T_b$)-SGC model: (a) external validation, (b) internal validation, (c) error distributions of the LOO-CV and ACGC models, and (d) $Y$-random validation

Figure 5 illustrates the error distributions of the ln$P_s$($T$, $T_b$)-ACGC and ln$P_s$($T$, $T_b$)-SGC models on the training set. The ln$P_s$($T$, $T_b$)-SGC model demonstrates significantly tighter error bounds, with errors (in ln(kPa)) concentrated within (-0.81, 0.96), compared to the broader range of (-0.88, 1.23) for the ln$P_s$($T$, $T_b$)-ACGC model. These findings clearly demonstrate the substantial advantage of incorporating molecular structural information for property estimation. Particularly, the superior performance of the SGC model highlights that accounting for molecular spatial specificity (stereo-structure) provides greater predictive accuracy than considering only topological features (as in the ACGC model). This compelling evidence underscores the critical importance of incorporating molecular three-dimensional structure for accurate property prediction. Calculation results show that the ACGC method can achieve satisfactory results even at low computational costs. Therefore, for different application scenarios and resource conditions, the most suitable solution can be selected between these two models.

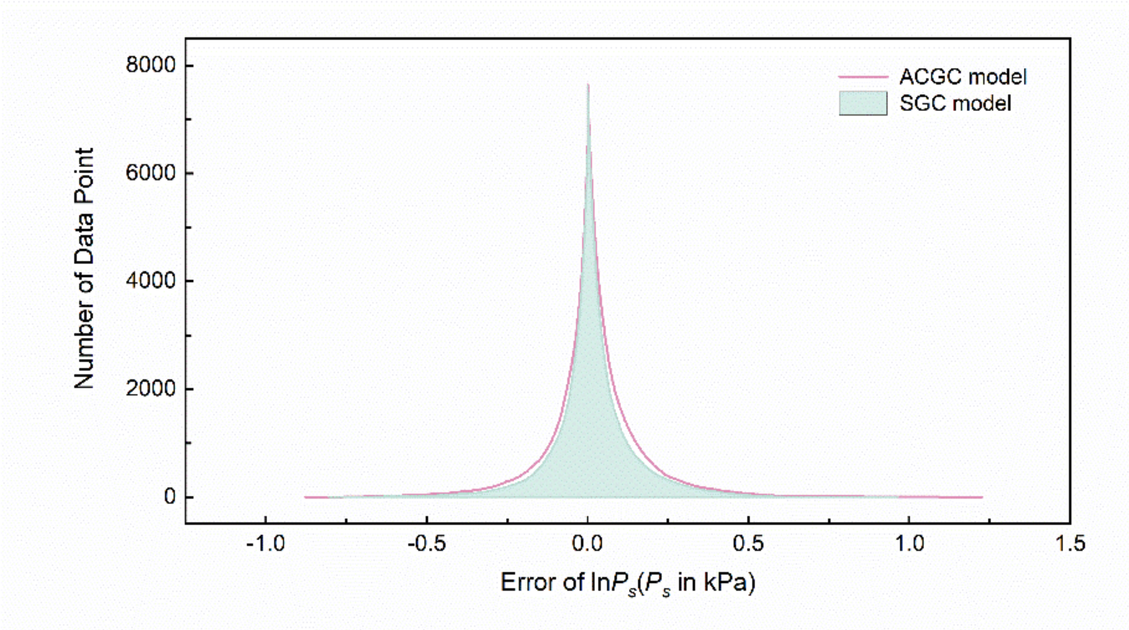


Figure 5. The error distributions of ln$P_s$($T$, $T_b$)-ACGC and ln$P_s$($T$, $T_b$)-SGC models

## 3.2. Effect of $T_b$ on models

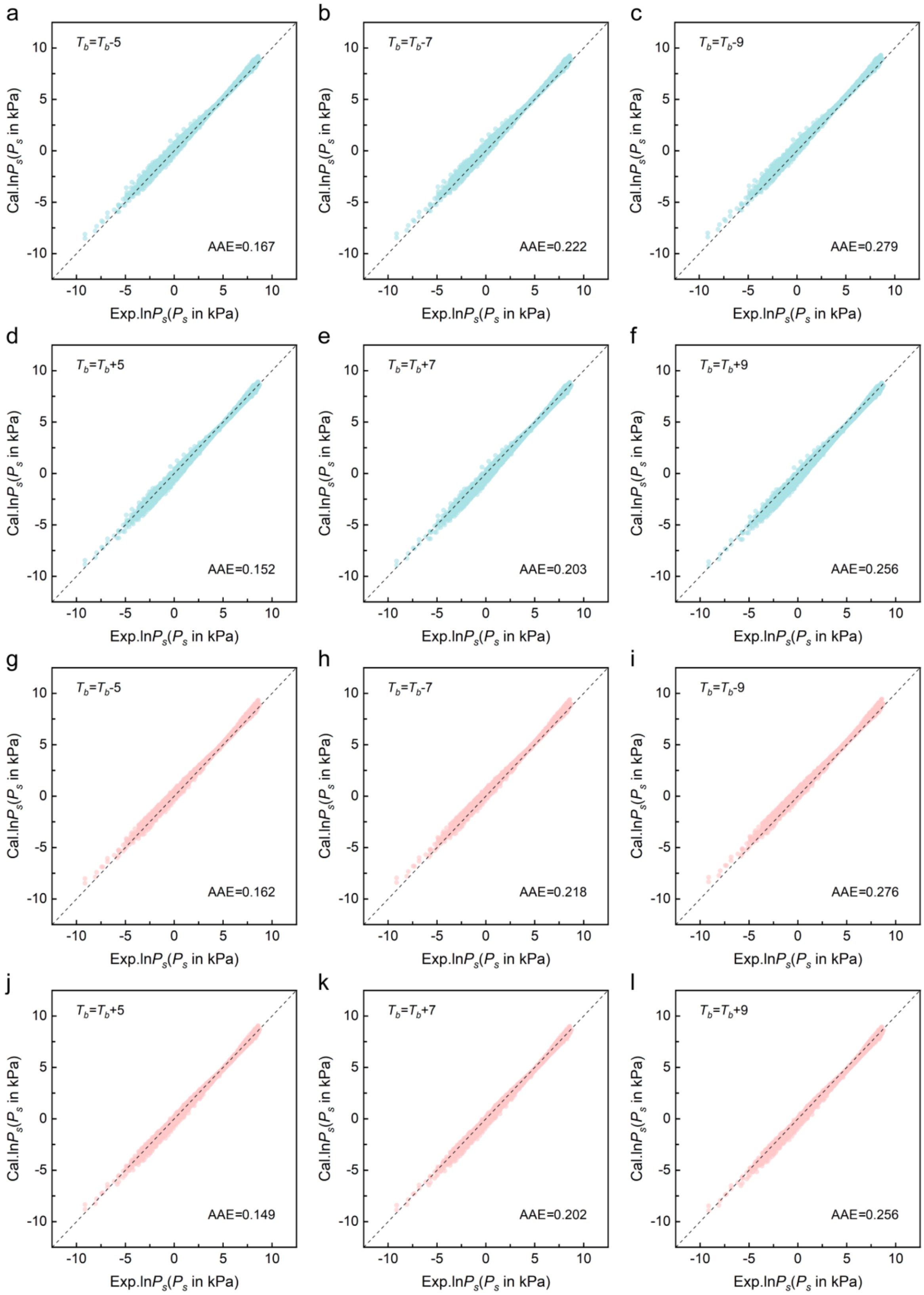


Figure 6. The calculated and experimental results after adjusting $T_b$: Light green represents the ln$P_s$($T$, $T_b$)-ACGC model (a~f), and pink represents the ln$P_s$($T$, $T_b$)-SGC model (g~l)

To investigate the influence of $T_b$ on the predictive performance of models, this study applied both positive and negative deviations to $T_b$. Figure 6 presents a comparison between the calculated values and experimental results after adjusting $T_b$. The analysis reveals that when the $T_b$ increases, the predicted values exhibit an overall downward shift, whereas a decrease in $T_b$ leads to an upward trend in predictions. This observation confirms a negative correlation between $T_b$ and predicted values, which aligns well with the model formulas (Eqs. (7) and (8)) described earlier.

Table 3. The AAE of training sets of models under different $T_b$ deviations

| $T_b+m$ | AAE (ln(kPa)) | | $T_b+m$ | AAE (ln(kPa)) | |
|---|---|---|---|---|---|
| | $\ln P_s(T, T_b)$-ACGC | $\ln P_s(T, T_b)$-SGC | | $\ln P_s(T, T_b)$-ACGC | $\ln P_s(T, T_b)$-SGC |
| $T_b$-0 | 0.067 | 0.057 | $T_b$+0 | 0.067 | 0.057 |
| $T_b$-5 | 0.167 | 0.162 | $T_b$+5 | 0.152 | 0.149 |
| $T_b$-6 | 0.194 | 0.190 | $T_b$+6 | 0.177 | 0.175 |
| $T_b$-7 | 0.222 | 0.218 | $T_b$+7 | 0.203 | 0.202 |
| $T_b$-8 | 0.250 | 0.247 | $T_b$+8 | 0.229 | 0.229 |
| $T_b$-9 | 0.279 | 0.276 | $T_b$+9 | 0.256 | 0.256 |
| $T_b$-10 | 0.309 | 0.306 | $T_b$+10 | 0.284 | 0.283 |

Table 3 summarizes the AAE of training sets of the models under different $T_b$ deviations. The results demonstrate that for both the $\ln P_s(T, T_b)$-ACGC and $\ln P_s(T, T_b)$-SGC models, a decrease in $T_b$ has a more pronounced impact on AAE. Furthermore, as the magnitude of $T_b$ deviation increases, the model errors show a clear upward trend. Further calculations (as provided in Table S3 of the Supporting Information) indicate that when the $T_b$ positive error reaches 12 K, the performance of the $\ln P_s(T, T_b)$-SGC model begins to fall below that of the $\ln P_s(T, T_b)$-ACGC model. Therefore, when the $T_b$ positive error is greater than or equal to 12 K, it is recommended to adopt the $\ln P_s(T, T_b)$-ACGC model. For other cases, the appropriate model should be selected based on specific application conditions and requirements. Additionally, according to literature reports, the QSPR model without $T_b$ developed by Feng et al.[31] has an AAE of 0.272 (ln(kPa)). As shown in Table 3, which lists the AAE results of two models within a $T_b$ offset range of 5~10 K, the models proposed in this study demonstrate better performance than the QSPR model without $T_b$ proposed by Feng et al.[31] when the $T_b$ error is controlled within 9 K.

### 3.3. Comparison results with existing GC methods

Table 4 presents a comparison between the $P_s$ models established in this study and other classical group contribution models. In existing research, Gharagheizi et al.[5] employed an artificial neural network-group contribution method to model 42000 vapor pressure data points for 1405 compounds, covering a temperature range of 55~3040 K, achieving an $R^2$ of 0.994. On the other hand, Naef and Acree[36] used a group additivity method to calculate the vapor pressure of 1907 organic molecules at 298.15 K. While their model achieved a high $R^2$ of 0.9945, it is only applicable to vapor pressure predictions at a single temperature (298.15 K), significantly limiting its practical utility. In contrast, the ACGC ($R^2$ = 0.9982) and SGC ($R^2$ = 0.9987) models developed in this study using linear regression methods demonstrated excellent predictive performance. Notably, the number of groups defined by atomic connectivity in our models (i.e., 77) are significantly fewer than those in the models by Gharagheizi et al.[5] (i.e., 202) and Naef and Acree[36] (i.e., 177). The comparative analysis shows that (1) incorporating both $T$ and $T_b$ into the modeling process improves the prediction accuracy of $P_s$; (2) under the same dataset conditions, selecting spatial structural features of molecules is more advantageous than topological structural features.

Table 4. Comparisons of the results for the $P_s$ models with those of other references

| References | Methods | Dataset | Range of $T$ | Number | $R^2$ |
|---|---|---|---|---|---|

| | | Compounds | Datapoints | (K) | of groups | |
|---|---|---|---|---|---|---|
| Gharagheizi et al.[5] | GC | 1405 | 42000 | 55~3040 | 202 | 0.994 |
| Naef and Acree[36] | GC | 1907 | 1907 | 298.15 | 177 | 0.9945 |
| ln$P_s$($T$, $T_b$)-ACGC | ACGC | 1062 | 17800 | 170~974 | 77 | 0.9982 |
| ln$P_s$($T$, $T_b$)-SGC | SGC | 1062 | 17800 | 170~974 | 77 | 0.9987 |

## 4. Conclusions

This study successfully employs a novel GC method integrating the topological and spatial structural features of molecules to construct the ln$P_s$($T$, $T_b$)-ACGC and ln$P_s$($T$, $T_b$)-SGC prediction models, with $T_b$ as a key parameter. The model performance evaluation results show that the ln$P_s$($T$, $T_b$)-ACGC model performs excellently on the training set, with an $R^2_{\text{training}}$ of 0.9982 and an $\text{AAE}_{\text{training}}$ of 0.067 ln(kPa). The ln$P_s$($T$, $T_b$)-SGC model demonstrates even better performance, achieving an $R^2_{\text{training}}$ of 0.9987 and an $\text{AAE}_{\text{training}}$ of 0.057 ln(kPa). In terms of model validation, the ln$P_s$($T$, $T_b$)-ACGC model exhibits strong predictive performance and stability, with external validation ($R^2_{\text{test}}$ = 0.9982) and internal validation ($Q^2_{\text{LOO}}$ = 0.9972). Similarly, the ln$P_s$($T$, $T_b$)-SGC model shows outstanding results, with external validation ($R^2_{\text{test}}$ = 0.9987) and internal validation ($Q^2_{\text{LOO}}$ = 0.9981), confirming its reliability and stability. Further analysis shows that when $T_b$ exhibits a positive offset greater than or equal to 12 K, the ln$P_s$($T$, $T_b$)-ACGC model demonstrates relatively better predictive performance.

In short, both the ln$P_s$($T$, $T_b$)-ACGC and ln$P_s$($T$, $T_b$)-SGC models established in this study can achieve high-precision $P_s$ predictions. Among them, the ln$P_s$($T$, $T_b$)-ACGC model can calculate $P_s$ through the topology, and the ln$P_s$($T$, $T_b$)-SGC model can achieve higher accuracy prediction due to the full consideration of the spatial structural features of molecules.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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